

7331__mnish__ass8

March 17, 2026

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[1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv('/home/csl-4/Downloads/titanics.csv')
df.head()
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[1]:
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	PassengerId	Survived	Pclass	\	Name	Sex	Age	SibSp	\
0	1	0	3		Braund, Mr. Owen Harris	male	22.0	1	
1	2	1	1		Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	3	1	3		Heikkinen, Miss. Laina	female	26.0	0	
3	4	1	1		Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	5	0	3		Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

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[2]: df.describe()
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[2]:
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	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	
std	257.353842	0.486592	0.836071	14.526497	1.102743	
min	1.000000	0.000000	1.000000	0.420000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	

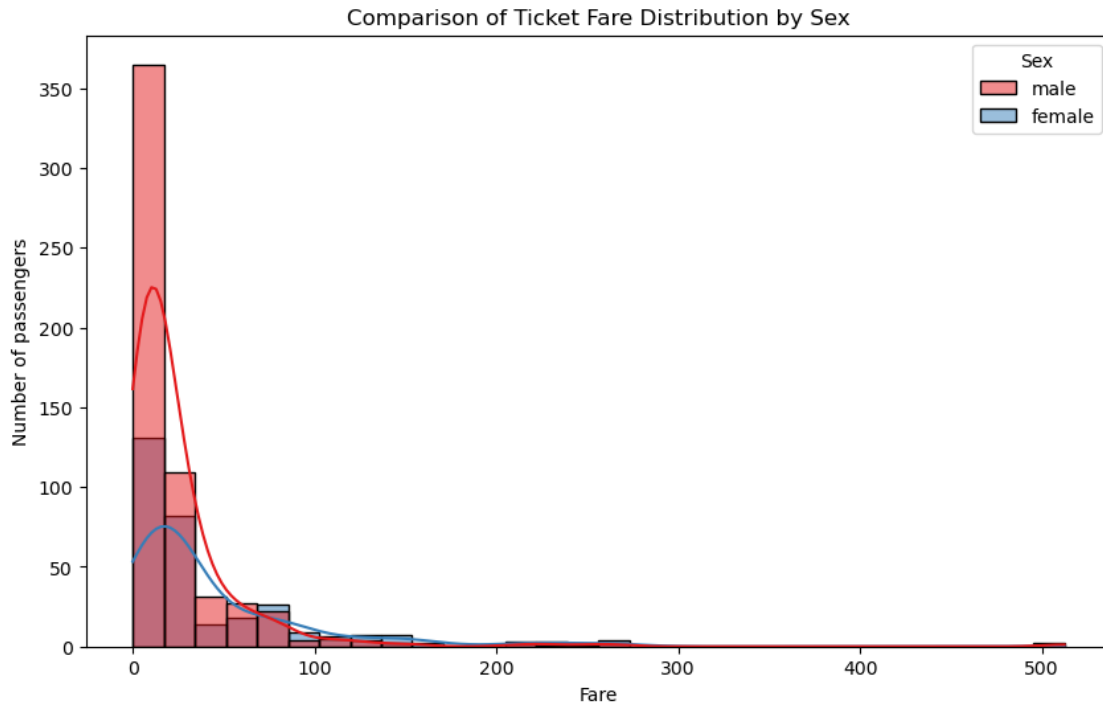
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

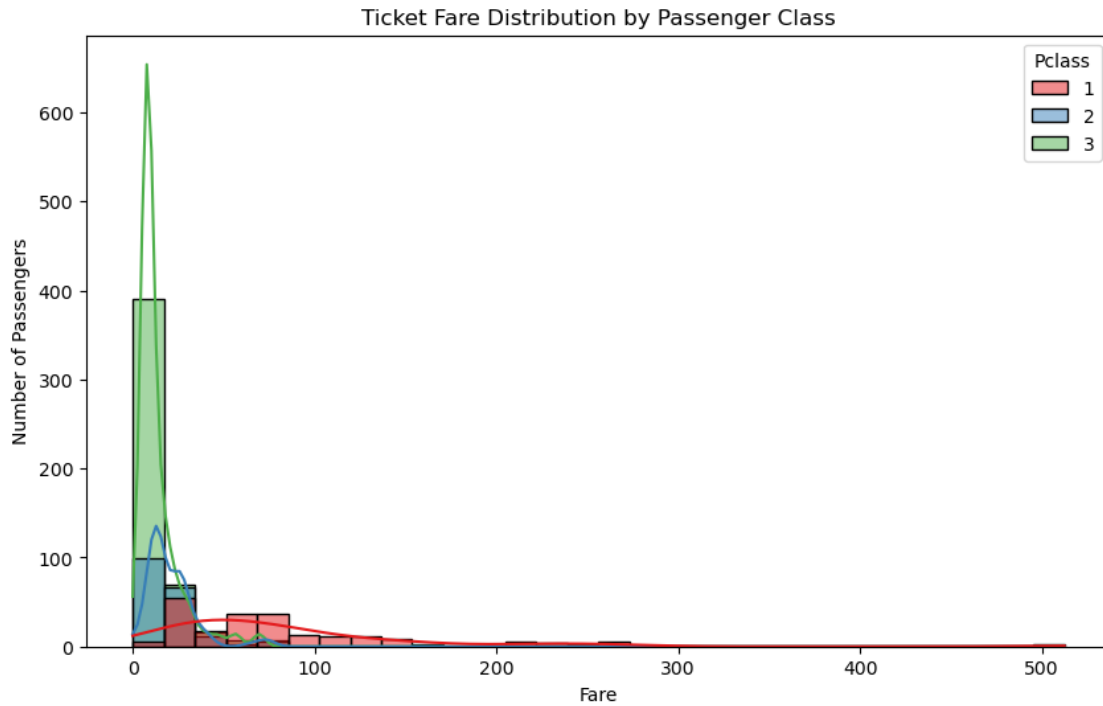
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[3]: df.info()
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<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null   int64
1   Survived        891 non-null   int64
2   Pclass          891 non-null   int64
3   Name            891 non-null   object
4   Sex             891 non-null   object
5   Age             714 non-null   float64
6   SibSp           891 non-null   int64
7   Parch           891 non-null   int64
8   Ticket          891 non-null   object
9   Fare            891 non-null   float64
10  Cabin           204 non-null   object
11  Embarked        889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
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[4]: plt.figure(figsize=(10,6))
sns.histplot(data=df, x='Fare', hue='Sex', bins=30, kde=True, palette='Set1')
plt.title('Comparison of Ticket Fare Distribution by Sex')
plt.xlabel('Fare')
plt.ylabel('Number of passengers')
plt.show()
```

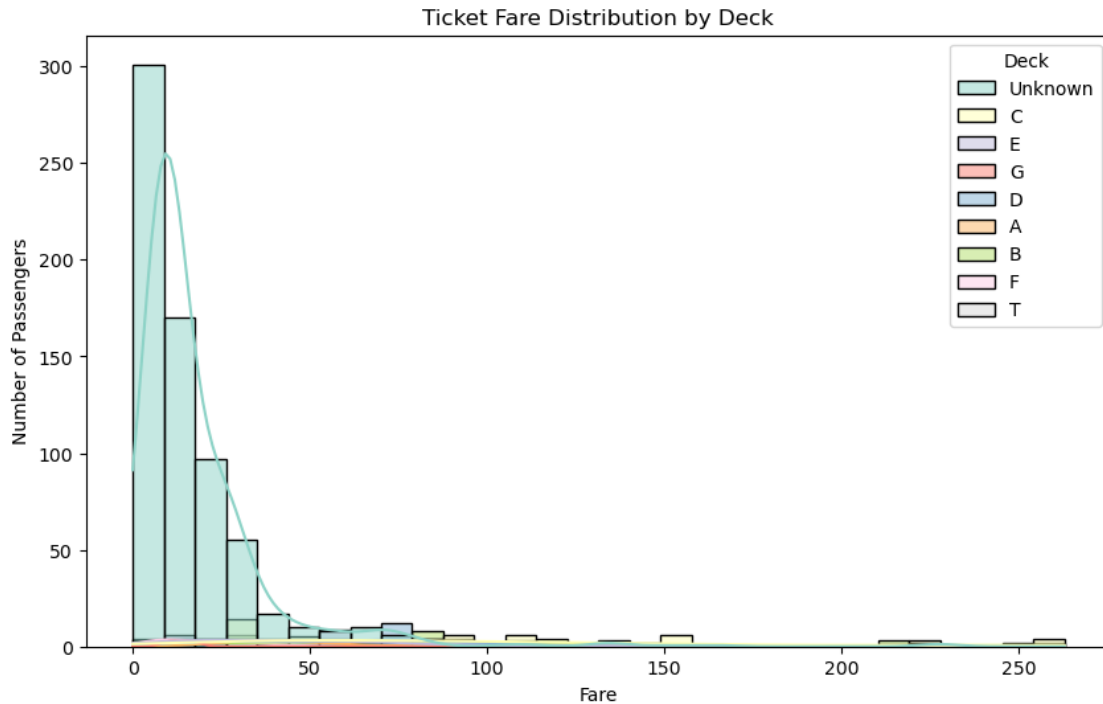


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[5]: plt.figure(figsize=(10,6))
sns.histplot(data=df, x='Fare', hue='Pclass', bins=30, kde=True, palette='Set1')
plt.title('Ticket Fare Distribution by Passenger Class')
plt.xlabel('Fare')
plt.ylabel('Number of Passengers')
plt.show()
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[6]: df['Cabin'] = df['Cabin'].fillna('Unknown')
df['Deck'] = df['Cabin'].apply(lambda x: x[0] if x != 'Unknown' else 'Unknown')
df_hist = df[df['Fare'] < 500]

plt.figure(figsize=(10,6))
sns.histplot(data=df_hist, x='Fare', hue='Deck', bins=30, kde=True,
             palette='Set3')
plt.title('Ticket Fare Distribution by Deck')
plt.xlabel('Fare')
plt.ylabel('Number of Passengers')
plt.show()
```



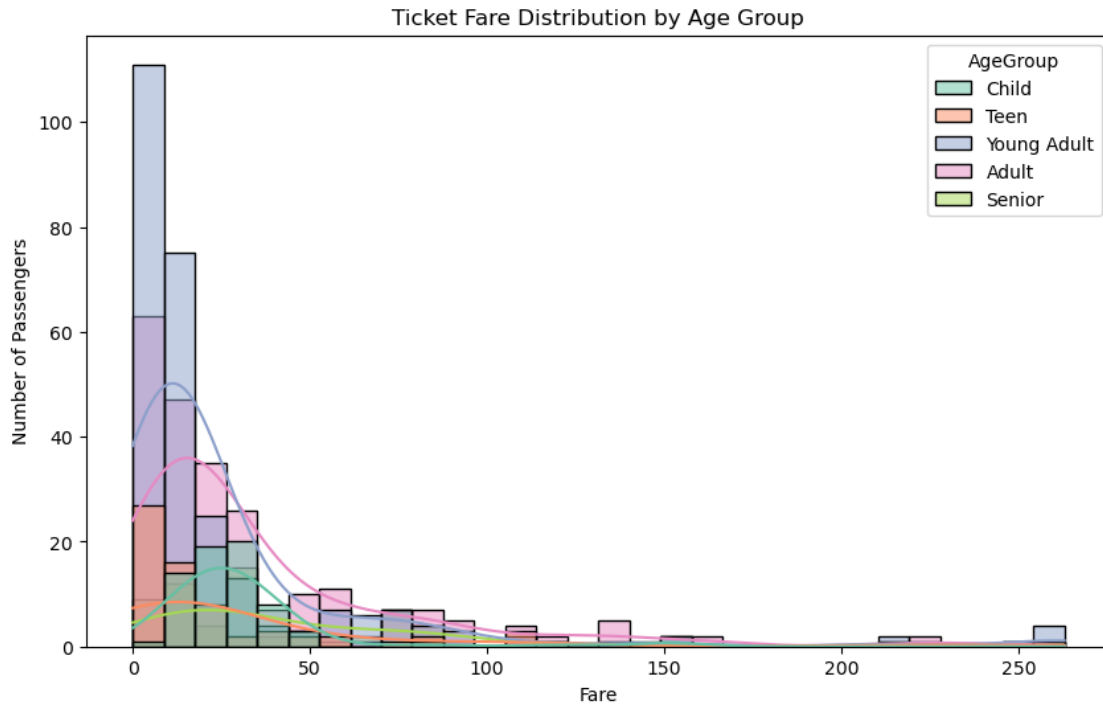
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[7]: print(df.columns)
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Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
      'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked', 'Deck'],
      dtype='object')
```

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[8]: df_clean = df.dropna(subset=['Age', 'Fare'])
df_clean = df_clean[df_clean['Fare'] < 500]

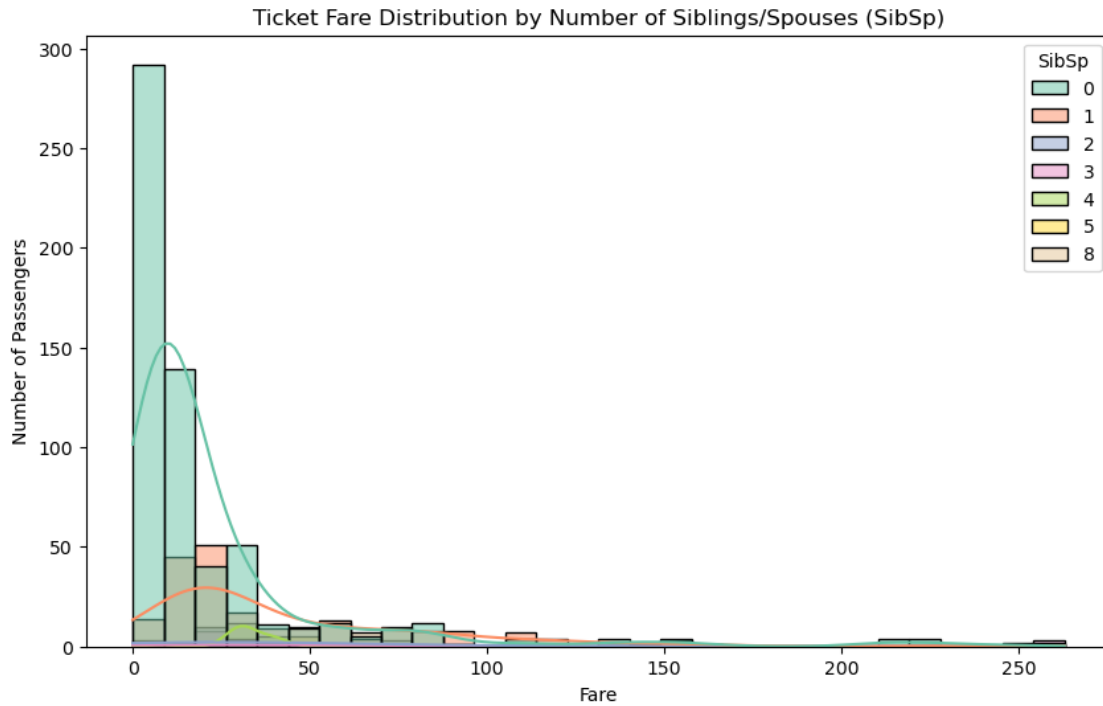
df_clean['AgeGroup'] = pd.cut(df_clean['Age'], bins=[0,12,18,30,50,80],
    ↳ labels=['Child', 'Teen', 'Young Adult', 'Adult', 'Senior'])

plt.figure(figsize=(10,6))
sns.histplot(data=df_clean, x='Fare', hue='AgeGroup', bins=30, kde=True,
    ↳ palette='Set2')
plt.title('Ticket Fare Distribution by Age Group')
plt.xlabel('Fare')
plt.ylabel('Number of Passengers')
plt.show()
```

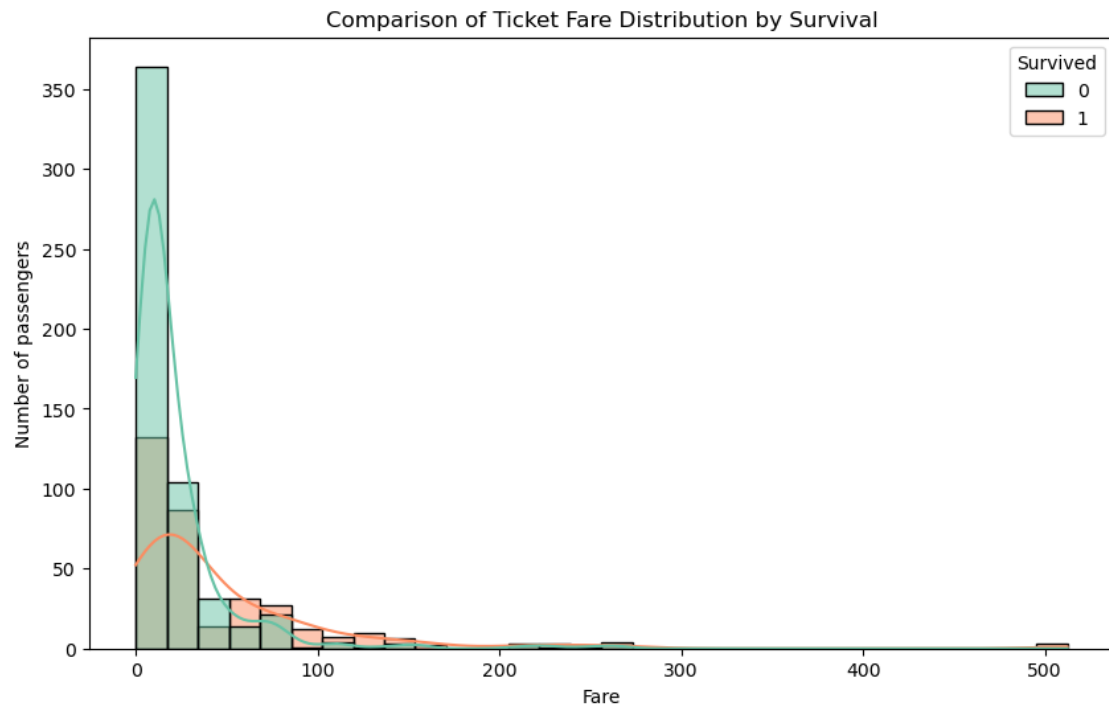


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[9]: df_clean = df[df['Fare'] < 500]

plt.figure(figsize=(10,6))
sns.histplot(data=df_clean, x='Fare', hue='SibSp', bins=30, kde=True,
             palette='Set2')
plt.title('Ticket Fare Distribution by Number of Siblings/Spouses (SibSp)')
plt.xlabel('Fare')
plt.ylabel('Number of Passengers')
plt.show()
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[15]: plt.figure(figsize=(10,6))
sns.histplot(data=df, x='Fare', hue='Survived', bins=30, kde=True,
             palette='Set2')
plt.title('Comparison of Ticket Fare Distribution by Survival')
plt.xlabel('Fare')
plt.ylabel('Number of passengers')
plt.show()
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